

Machine Learning Driven Temporal Sub-Phenotypes of Early Sepsis

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Introduction

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection. It remains a major global health challenge with approximately **49 million cases and 11 million deaths annually**[1]. Sepsis can lead to tissue damage, organ failure, and death if not treated promptly.

Traditional clinical frameworks often treat sepsis as a single disease entity using metrics such as **SIRS** and **SOFA scores** [2]. However, sepsis patients exhibit significant heterogeneity in clinical presentation and disease progression. Identifying distinct patient **sub-phenotypes** can enable personalized treatment strategies and improve clinical outcomes.

Dataset

The study uses the **MIMIC-IV** [3] a publicly available dataset containing de-identified clinical records from ICU patients between 2008 and 2019.

The dataset includes:

- Demographics
- Vital signs
- Laboratory measurements
- Medication records

Cohort Selection Criteria

- Age ≥ 18 years
- First ICU admission
- Patients surviving at least 24 hours after ICU admission

Acknowledgements



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Motivation and Objective

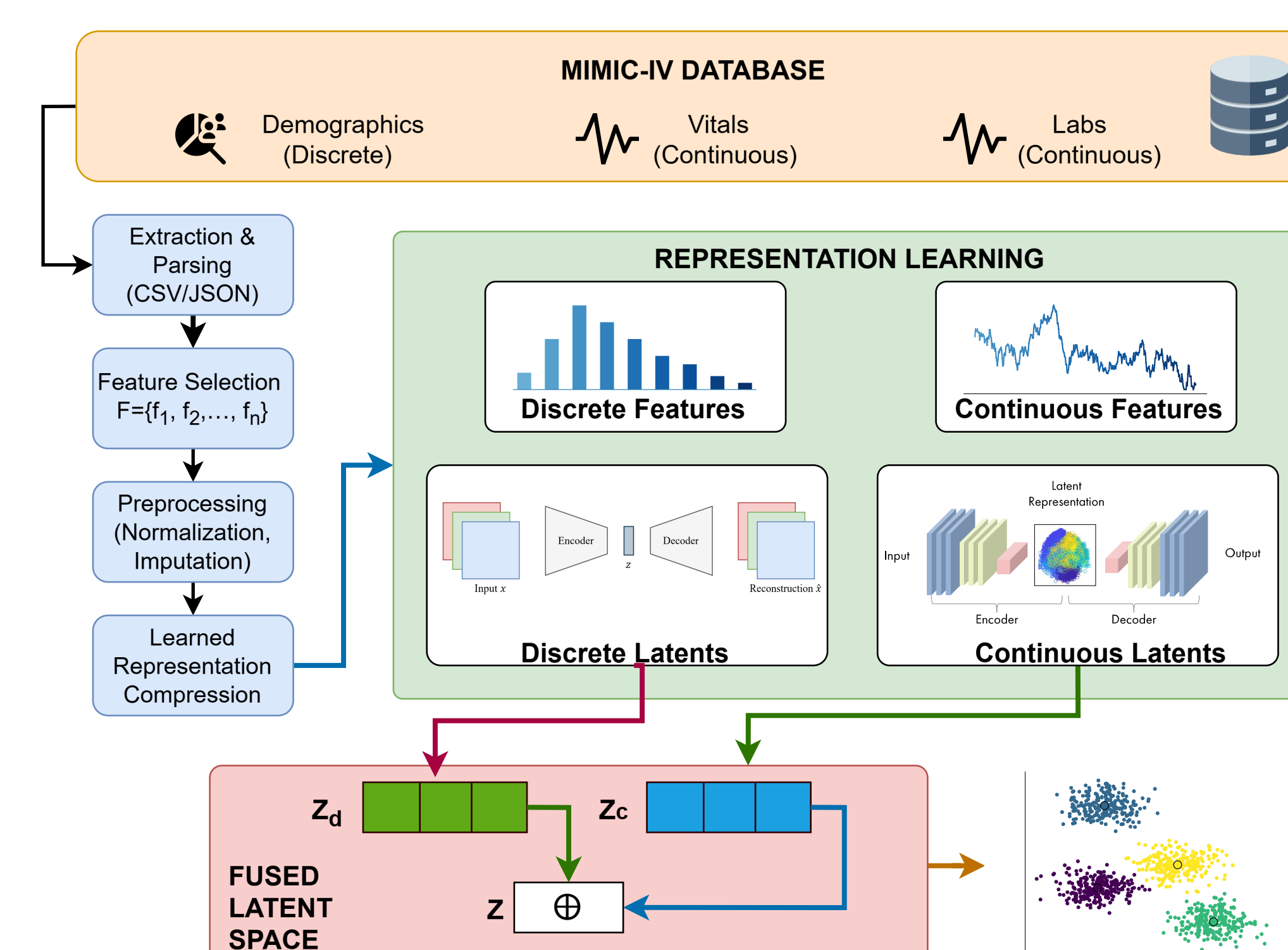
Existing machine learning approaches for sepsis phenotyping have several limitations.

- Many studies rely on **static snapshot data** rather than temporal patient trajectories.
- Temporal disease progression during early ICU admission is often ignored.
- Current models may fail to capture the dynamic evolution of sepsis.

This study identifies **temporal sub-phenotypes of early sepsis**

- Analyze patient trajectories during the **first 24 hours of ICU admission**.
- Integrate **static clinical variables** and **sequential physiological signals**.
- Learn compact representations using **autoencoder architectures**.
- Discover distinct patient groups using **clustering techniques**.
- Predict patient sub-phenotypes using **temporal machine learning models**.

Pipeline



Methodology

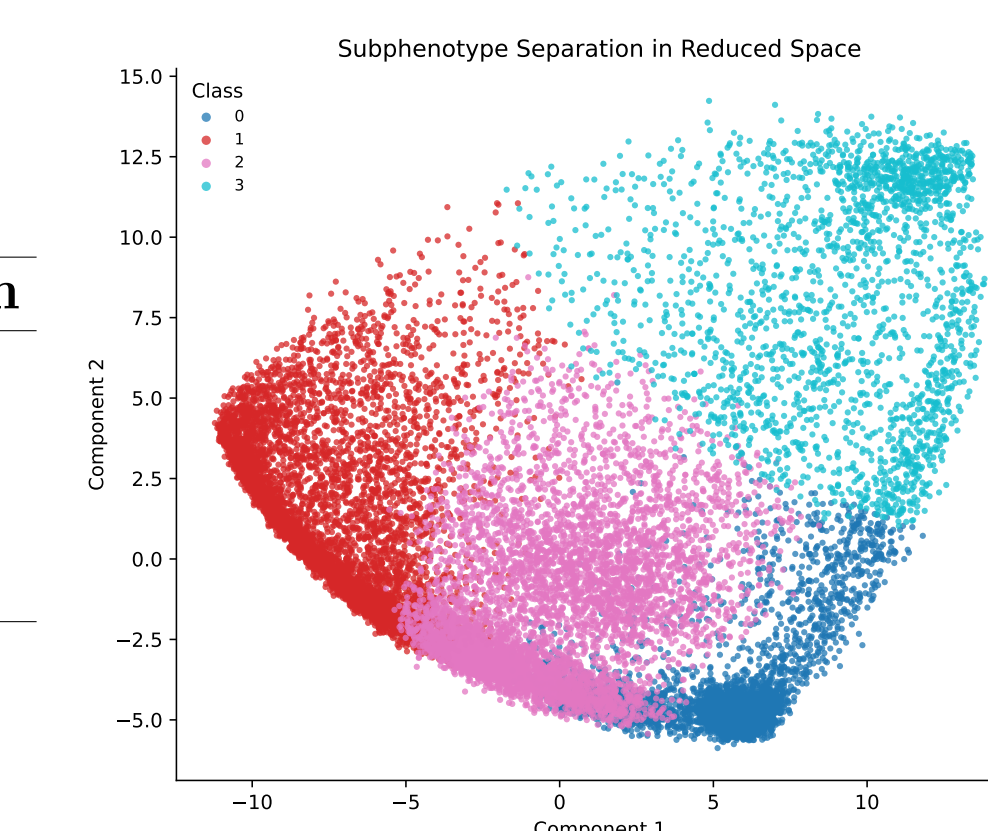
The framework integrates representation learning and clustering to identify meaningful patient phenotypes.

- Static clinical features are encoded using an **MLP Autoencoder**.
- Sequential physiological signals are encoded using an **LSTM Autoencoder**.
- Latent representations are concatenated to create a unified patient representation.
- **K-Means clustering** is applied to identify temporal sub-phenotypes.

Clustering Results

Four temporal sub-phenotypes were identified using **K-Means clustering** on the learned latent representations.

Subphenotypes	Proportion
SP1	46%
SP2	29%
SP3	11%
SP4	14%



Outcomes Across Sub-Phenotypes

Clinical outcomes differed significantly across the discovered sub-phenotypes.

Outcome	SP1	SP2	SP3	SP4
ICU stay time	111.41	85.44	174.34	183.20
Hospital stay time	280.38	250.16	342.98	363.21
Mortality rate (%)	9.42	8.67	16.86	29.70

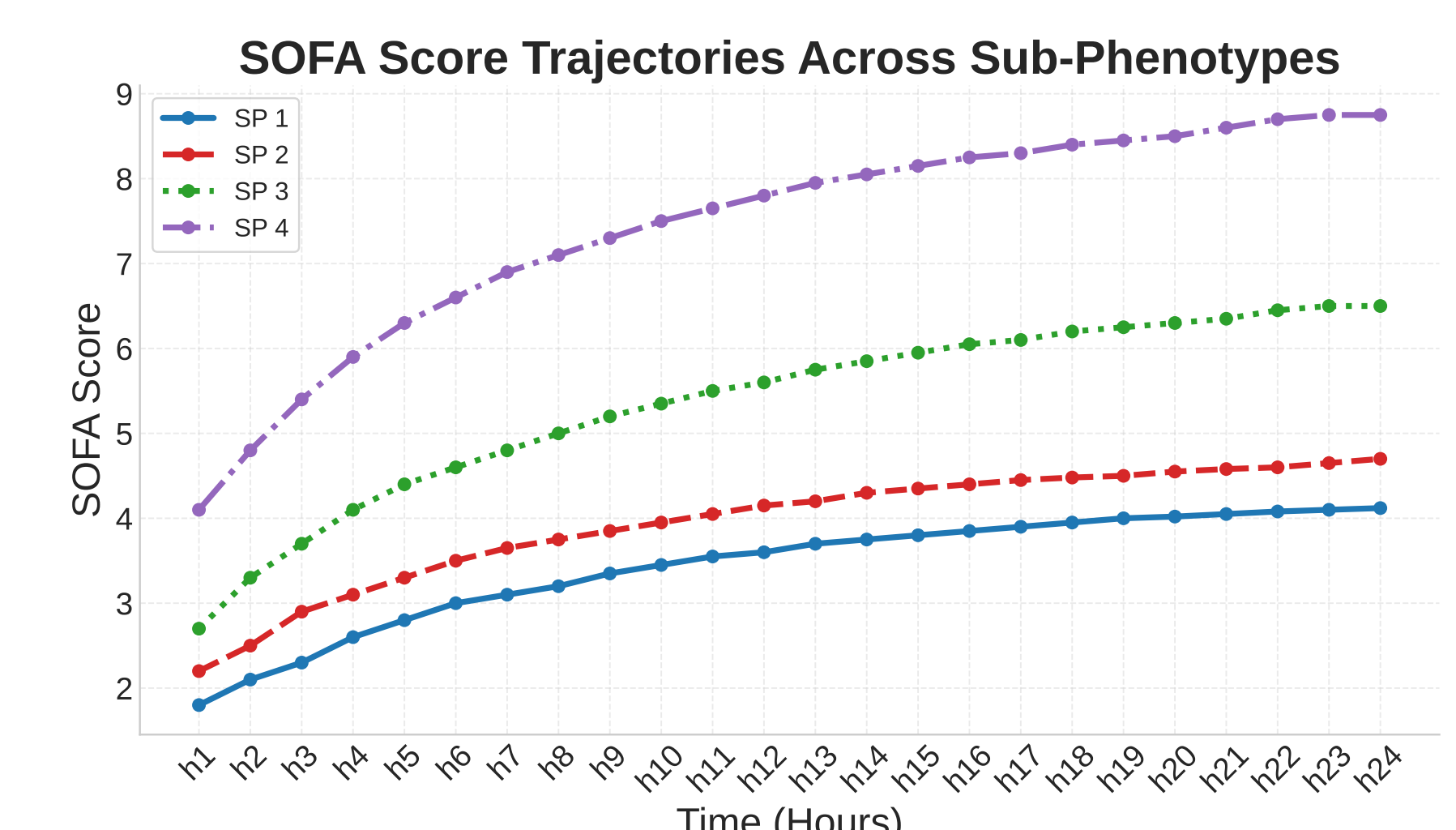
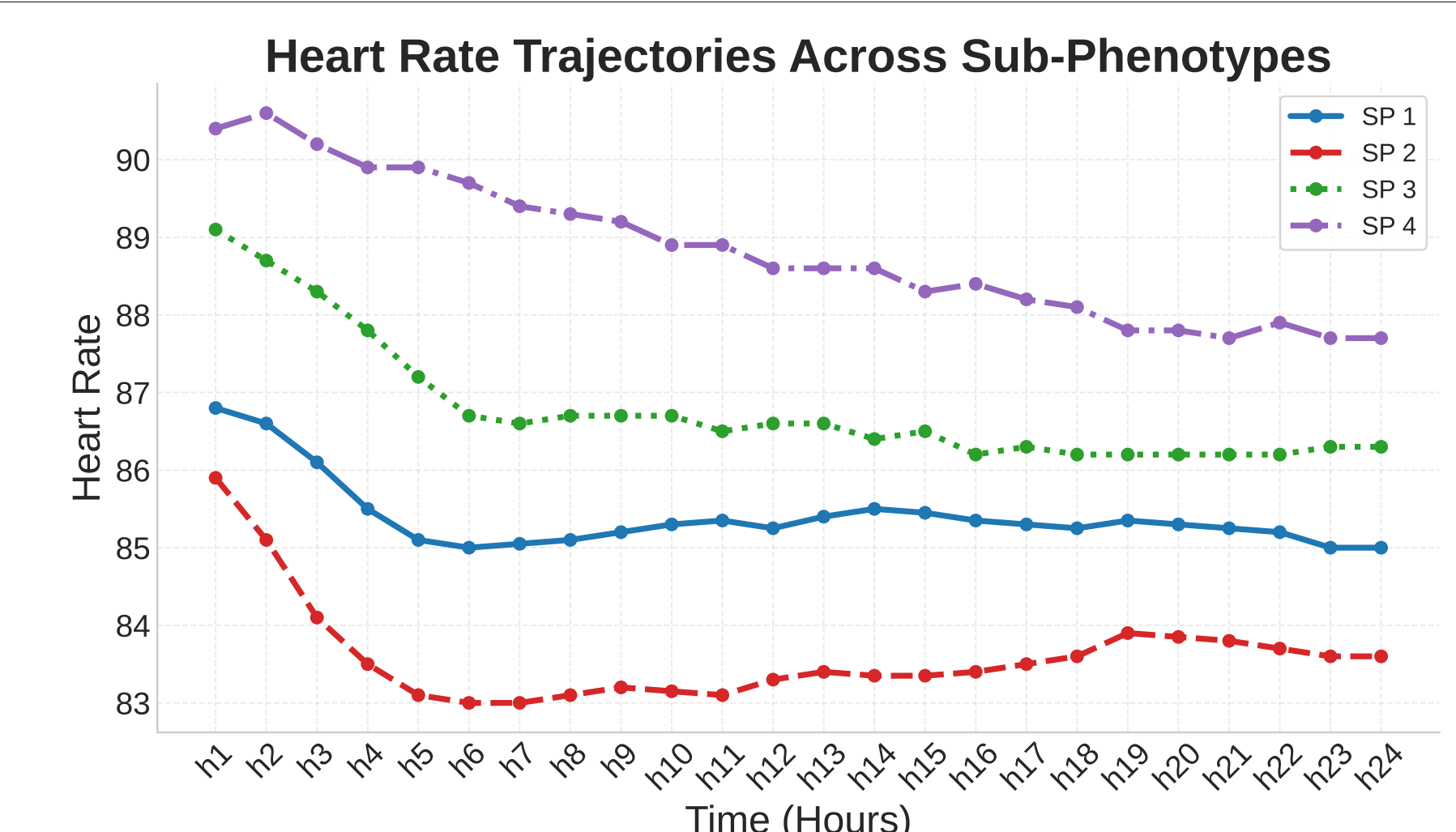
- SP2 showed the lowest ICU and hospital stay durations
- SP4 showed the longest ICU and hospital stay durations
- SP4 had the highest mortality rate: **29.70%**

Clinical Characteristics of Sub-Phenotypes

Sub-phenotypes show clear separation in disease severity. SP3 and SP4 exhibit higher SOFA scores, increased vasopressor use, and longer ventilation time compared to SP1 and SP2, indicating more severe organ dysfunction.

Temporal trajectories reinforce this pattern. SP3 and SP4 maintain persistently elevated SOFA scores over the first 24 hours, while SP1 and SP2 remain stable, highlighting distinct progression dynamics in early sepsis.

Recurrent Features	SP1	SP2	SP3	SP4
Vent time	35.24 $\pm$ 7.99	23.74 $\pm$ 6.38	60.19 $\pm$ 14.84	70.52 $\pm$ 11.58
SOFA	3.57 $\pm$ 0.90	4.09 $\pm$ 0.87	5.96 $\pm$ 1.54	7.57 $\pm$ 1.58
Vasopressor	0.03 $\pm$ 0.04	0.05 $\pm$ 0.06	1.07 $\pm$ 0.60	1.92 $\pm$ 0.96



References

- [1] K. Rudd. Global sepsis burden. *Lancet*, 2020.
- [2] M. Singer. Sepsis-3. *JAMA*, 2016.
- [3] A. Johnson. MIMIC-IV. *PhysioNet*, 2023.